

# Wenfan Wei

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## TECHNICAL SKILLS

**Programming:** Python, C++, C, CUDA, x86 Assembly, Java, SQL

**Robotics & Simulation:** Isaac Sim, Isaac Lab, ROS 2, SLAM

**Machine Learning & Computer Vision:** PyTorch, TensorFlow, NumPy, pandas

**Developer Tools:** Git, Docker, MATLAB, Linux, AWS

## EXPERIENCE

### AMR Automation Intern

August 2026 – Present

*Wellwit Robotics*

*Los Angeles, CA*

- Fine-tuning and validating onboard AMR sensors and perception parameters to improve data quality for localization and navigation.
- Working with ROS 2 and SLAM pipelines to build maps, evaluate localization, and troubleshoot navigation behavior while assisting with telemetry collection, issue reproduction, and system-parameter tuning.

### Robotics Intern

May 2026 – August 2026

*MAXIEYE*

*Shanghai, China*

- Built an NVIDIA Isaac Sim synthetic data generation pipeline for robotic tube grasping with domain randomization, exporting RGB, depth, segmentation, and bounding-box annotations.
- Generated and validated over 2,000 robot demonstrations totaling 350,000 synchronized frames with RGB camera footage, 7-DOF joint states, and action targets.
- Fine-tuned YOLO11 segmentation and pi0.5 VLA models on simulated and teleoperated data using Colab and physical GPUs, and conducted reinforcement learning in Isaac Lab for sim-to-real deployment.
- Aligned RGB segmentation masks with depth, back-projected pixels into 3D point clouds, estimated tube centers, and generated grasp candidates ranked by IK feasibility and collision clearance.
- Developed a humanoid robot trajectory-cleaning pipeline using spike removal, Savitzky–Golay and One Euro filtering, with forward-kinematics validation to reduce motion noise while preserving end-effector accuracy.

### Software Intern

June 2025 – August 2025

*Amazon Web Services Cal Poly SLO DxHub*

*San Luis Obispo, CA*

- Selected for an 80-person internship cohort from more than 1,400 applicants.
- Built a production-ready NumPy and pandas pipeline to clean and normalize more than 5,000 institutional receipt records.
- Developed an AWS-hosted RAG agent and custom interface to automate receipt auditing and flag policy violations; presented results to Amazon engineers and university stakeholders.

## PROJECTS

### NormaCore Robot Arm

- Awarded NormaCore Student Grant including NormaCore ElRobot Arm and cloud compute units for background in robotics.
- Built, wired, configured, and calibrated 7-dof robot arm from 3D-printed components and servo motors, fine-tuned actuators for teleoperation.
- Implemented inverse-kinematics solvers and grasp-point planning to generate accurate robot trajectories for VLA policy training.

### UAV Geospatial Target Fusion

- Built an Isaac Sim synthetic-perception pipeline to generate UAV imagery with randomized camera poses and environmental conditions, exporting datasets for YOLO object-detection training.
- Developing a multi-view target-localization pipeline that projects 2D detections into 3D rays using camera intrinsics and poses, then fuses observations to estimate target locations.

### Libreye Eagle

- Built a privacy-focused security camera system using Frigate NVR, Docker, and a fine-tuned YOLO11 object-detection model.
- Designed and 3D-printed a thermally managed enclosure and assembled camera modules with power controls, microSD storage, and battery support.
- Configured FFmpeg, RTSP, and Ethernet streaming on a home server for low-latency video processing and local data ownership.

## EDUCATION

### University of California, Irvine

Irvine, CA

*Bachelor of Science, Computer Engineering*

*Expected July 2028*

- **Activities:** Financial Officer, ZOTBotics, UAV Forge Perception Team

### De Anza College

Cupertino, CA

*Associate of Science, Computer Engineering | GPA: 3.6*

*June 2024 – June 2026*

- **Coursework:** Data Structures, Computer Architecture and x86 Assembly, Circuits, Differential Equations, Linear Algebra, and Physics
- **Activities:** President, Mathematical Investment Trading Society; Treasurer, Developers Guild; Project Lead, Foothill Computer Engineering Club